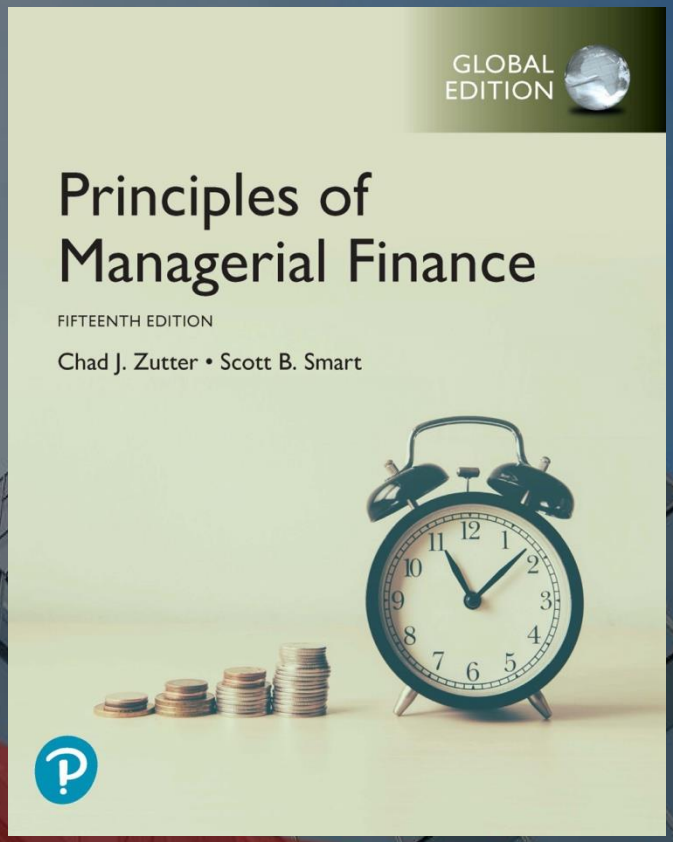
Week 3

Long- and Short-Term Financial Planning



Personal Finance Example 4.1 (1 of 2)

The first step in personal financial planning requires you to define your goals. Whereas in a corporation the goal is to maximize shareholder wealth, individuals typically have several different financial goals. These might have short-, medium-, or long-term horizons, and they depend on your age, income, family status, and other factors.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Personal Finance Example 4.1 (2 of 2)

Set your personal financial goals carefully and realistically. Each goal should be clearly defined and have a priority, time frame, and cost estimate. For example, a college senior's intermediate-term goal in 2019 might include earning a master's degree at a cost of $60,000 by 2021, and his or her long-term goal might be to buy a condominium at a cost of $250,000 by 2025.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (1 of 11)

- Depreciation
  - A portion of the costs of fixed assets charged against annual revenues over time
    - Modified Accelerated Cost Recovery System (MACRS)
      - System used to determine the depreciation of assets for tax purposes
      - Prior to the Tax Cuts and Jobs Act of 2017, firms were required to depreciate most types of assets using MACRS
      - The new law allows firms to immediately deduct 100% of the cost of many (but not all) types of new assets
      - However, these "immediate" deductions will be phased out gradually over several years without further Congressional action

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (2 of 11)

- Depreciation
  - Depreciable Value of an Asset
    - Under the basic MACRS procedures, the depreciable value of an asset is its full cost, including outlays for installation

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.2

Baker Corporation acquired a new machine at a cost of $38,000, with installation costs of $2,000. When the machine is retired from service, Baker expects to sell it for scrap metal and receive $1,000. Regardless of its expected salvage value, the depreciable value of the machine is $40,000: $38,000 cost + $2,000 installation cost.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (3 of 11)

- Depreciation
  - Depreciable Life of an Asset
    - Time period over which an asset is depreciated
      - MACRS Recovery Period
        - The appropriate depreciable life of a particular asset as determined by MACRS
    - The Tax Cuts and Jobs Act of 2017 allows firms to depreciate 100% of an asset's cost in the year it is purchased, though some types of assets are not eligible for this bonus depreciation

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (4 of 11)

- Depreciation Methods
  - For financial reporting purposes, companies can use a variety of depreciation methods (straight-line, double-declining balance, and sum-of-the-years'-digits)
  - For tax purposes, firms usually must depreciate assets in the first four MACRS property classes by the double-declining balance method, using a half-year convention (meaning that a half-year's depreciation is taken in the year the asset is purchased) and switching to straight-line when advantageous

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.1 First Four Property Classes under MACRS

Property class (recovery period)	Definition
3 years	Research equipment and certain special tools
5 years	Computers, printers, copiers, duplicating equipment, cars, light-duty trucks, qualified technological equipment, and similar assets
7 years	Office furniture, fixtures, most manufacturing equipment, railroad track, and single-purpose agricultural and horticultural structures
10 years	Equipment used in petroleum refining or in the manufacture of tobacco products and certain food products

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.2 Rounded Depreciation Percentages by Recovery Year Using MACRS for First Four Property Classes

	Percentage by recovery year[a]			
Recovery year	**3 years**	**5 years**	**7 years**	**10 years**
1	33%	20%	14%	10%
2	45	32	25	18
3	15	19	18	14
4	7	12	12	12
5		12	9	9
6		5	9	8
7			9	7
8			4	6
9				6
10				6
11				4
Totals	100%	100%	100%	100%

[a] These percentages have been rounded to the nearest whole percent.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.3

Baker Corporation acquired, for an installed cost of $40,000, a machine having a recovery period of 5 years. Using the applicable percentages from Table 4.2, Baker calculates the depreciation in each year as follows:

Year	Cost	Percentages (from Table 4.2)	Depreciation
1	$40,000	20%	$40,000 × 20% = $8,000
2	40,000	32	$40,000 × 32% = $12,800
3	40,000	19	$40,000 × 19% = $7,600
4	40,000	12	$40,000 × 12% = $4,800
5	40,000	12	$40,000 × 12% = $4,800
6	40,000	5	$40,000 × 5% = $2,000
Totals		100%	$40,000

Column 3 shows that the full cost of the asset is written off over 6 recovery years.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (5 of 11)

- Developing the Statement of Cash Flows
  - Cash Flow from Operating Activities
    - Cash flows directly related to sale and production of the firm's products and services
  - Cash Flow from Investment Activities
    - Cash flows associated with purchase and sale of both fixed assets and equity investments in other firms
  - Cash Flow from Financing Activities
    - Cash flows that result from debt and equity financing transactions; includes incurrence and repayment of debt, cash inflows from the sale of stock, and cash outflows to repurchase stock or pay cash dividends

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (6 of 11)

- Developing the Statement of Cash Flows
  - Classifying Inflows and Outflows of Cash
    - A decrease in an asset is an inflow of cash
    - An increase in an asset is an outflow of cash
    - A decrease in a liability is an outflow of cash
    - An increase in a liability is an inflow of cash

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.3 Inflows and Outflows of Cash

Inflows (sources)	Outflows (uses)
Decrease in any asset	Increase in any asset
Increase in any liability	Decrease in any liability
Net profit after taxes	Net loss after taxes
Depreciation and other noncash charges	Dividends paid
Sale of stock	Repurchase or retirement of stock

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (7 of 11)

- Developing the Statement of Cash Flows
  - Classifying Inflows and Outflows of Cash
    - Depreciation and amortization are noncash charges
      - Noncash Charge
        - An expense that is deducted on the income statement but does not involve the actual outlay of cash during the period; includes depreciation, amortization, and depletion
      - Only changes in gross net fixed assets appear on the statement of cash flows
      - Direct entries of changes in retained earnings are not included on the statement of cash flows

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (8 of 11)

- Developing the Statement of Cash Flows
  - Preparing the Statement of Cash Flows
    - All cash inflows as well as net profit after taxes and depreciation are treated as positive values
    - All cash outflows, any losses, and dividends paid are treated as negative values

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.4 Baker Corporation 2019 Income Statement ($000)

Sales revenue	$ 1,700
Less: Cost of goods sold	1,000
Gross profits	$ 700
Less: Operating expenses	
Selling, general, and administrative expense	$ 230
Depreciation expense	100
Total operating expense	$ 330
Earnings before interest and taxes (EBIT)	$ 370
Less: Interest expense	70
Net profit before taxes	$ 300
Less: Taxes (rate = 21%)	63
Net profit after taxes	$ 237
Less: Preferred stock dividends	10
Earnings available for common stockholders	$ 227
Earnings per share (EPS)[a]	$ 2.27
Common stock dividends per share (DPS)[b]	$ 0.70

[a]Calculated by dividing the earnings available for common stockholders by the number of shares of common stock outstanding ($170,000 ÷ 100,000 shares = $1.70 per share).
[b]The firm's board decided to pay $70,000 in dividends to common shareholders, so dividends per share are $0.70 ($70,000÷ 100,000 shares = $0.70 per share).

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.5 Baker Corporation Balance Sheets ($000) (1 of 2)

	December 31	
Assets	**2019**	**2018**
Cash and marketable securities	$ 1,057	$ 500
Accounts receivable	400	500
Inventories	600	900
Total current assets	$ 2,000	$1,900
Land and buildings	$ 1,200	$1,050
Machinery and equipment, furniture and fixtures, vehicles, and other	1,300	1,150
Total gross fixed assets (at cost)	$ 2,500	$2,200
Less: Accumulated depreciation	1,300	1,200
Net fixed assets	$ 1,200	$1,000
Total assets	$ 3,257	$2,900
Liabilities and stockholders' equity		
Accounts payable	$ 700	$ 500
Notes payable	600	700

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.5 Baker Corporation Balance Sheets ($000) (2 of 2)

	December 31	
Assets	**2019**	**2018**
Liabilities and stockholders' equity		
Accruals	100	200
Total current liabilities	$1,400	$1,400
Long-term debt	600	400
Total liabilities	$2,000	$1,800
Preferred stock	$ 100	$ 100
Common stock: $1.20 par, 100,000 shares outstanding in 2019 and 2018	120	120
Paid-in capital in excess of par on common stock	380	380
Retained earnings	657	500
Total stockholders' equity	$1,257	$1,100
Total liabilities and stockholders' equity	$3,257	$2,900

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.6 Baker Corporation Statement of Cash Flows ($000) for the Year Ended December 31, 2019 (1 of 2)

Cash flow from operating activities	
Net profit after taxes	$ 237
Depreciation	100
Decrease in accounts receivable	100
Decrease in inventories	300
Increase in accounts payable	200
Decrease in accruals	–100
Cash provided by operating activities	$837
Cash flow from investment activities	
Increase in gross fixed assets	–$300
Changes in equity investments in other firms	0
Cash provided by investment activities	–$300

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.6 Baker Corporation Statement of Cash Flows ($000) for the Year Ended December 31, 2019 (2 of 2)

Cash flow from financing activities	
Decrease in notes payable	–$ 100
Increase in long-term debt	200
Changes in stockholders' equity[a]	0
Dividends paid (common and preferred)	–80
Cash provided by financing activities	$ 20
Net increase in cash and marketable securities	$ 557

[a] Retained earnings are excluded here because their change is actually reflected in the combination of the "Net profit after taxes" and "Dividends paid" entries.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (9 of 11)

- Developing the Statement of Cash Flows
  - Interpreting the Statement
    - Operating Cash Flow (OCF)
      - The cash flow a firm generates from its normal operations; calculated as net operating profits after taxes (NOPAT) plus depreciation
      - Net Operating Profit After Taxes (NOPAT) = EBIT × (1−*T*)

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (10 of 11)

- Developing the Statement of Cash Flows
  - Interpreting the Statement
    - Operating Cash Flow (OCF)

$$\text{OCF} = \text{NOPAT} + \text{Depreciation} = [\text{EBIT} \times (1 - \text{T})] + \text{Depreciation} \qquad (4.3)$$

  - With the passage of the Tax Cuts and Jobs Act of 2017, the top marginal corporate tax rate fell from 35% to a flat 21%
  - For sole proprietors and partnerships, individual tax rates are applicable.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.4

Substituting the values for Baker Corporation from its income statement (Table 4.4) into Equation 4.3, we get

$$OCF = [\$370 \times (1.00 - 0.21)] + \$100 = \$292.3 + \$100 = \$392.3$$

During 2019, Baker Corporation generated $392,300 of cash flow from producing and selling its output. Therefore, we can conclude that Baker's operations are generating positive cash flows.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.2 Measuring the Firm's Cash Flow (11 of 11)

- Free Cash Flow (FCF)
  - The amount of cash flow available to investors (creditors and owners) after the firm has met all operating needs and paid for investments in net fixed assets and net current assets

$$\text{FCF} = \text{OCF} - \text{Net Fixed Asset Investment} - \text{Net Current Asset Investment} \quad (4.4)$$

  - Net Fixed Asset Investment = Change in Net Fixed Assets + Depreciation
  - Net Current Asset Investment = Change in Current Assets − Change in (Accounts Payable + Accruals)

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.5

Using Baker Corporation's balance sheets in Table 4.5, we see that its change in net fixed assets between 2018 and 2019 was $200 ($1,200 in 2019 – $1,000 in 2018). Substituting this value and the $100 of depreciation for 2019 into Equation 4.5, we get Baker's net fixed asset investment (NFAI) for 2019:

$$\text{NFAI} = \text{Change in net fixed assets} + \text{Depreciation} \qquad (4.5)$$

$$\text{NFAI} = \$200 + \$100 = \$300$$

Baker Corporation therefore invested a net $300,000 in fixed assets during 2019. This amount would, of course, represent a cash outflow to acquire fixed assets during 2019.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.6 (1 of 3)

Looking at Baker Corporation's balance sheets for 2018 and 2019 in Table 4.5, we see that the change in current assets between 2018 and 2019 is $157 ($2,057 in 2019 – $1,900 in 2018). The difference between Baker's accounts payable plus accruals of $800 in 2019 ($700 in accounts payable + $100 in accruals) and of $700 in 2018 ($500 in accounts payable + $200 in accruals) is $100 ($800 in 2019 – $700 in 2018). Substituting into Equation 4.6 the change in current assets and the change in the sum of accounts payable plus accruals for Baker Corporation, we get its 2019 NCAI:

$$\text{NCAI} = \text{Change in current assets} - \text{Change in (accounts payable} + \text{accruals)} \qquad (4.6)$$

$$\text{NCAI} = \$157 - \$100 = \$57$$

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.6 (2 of 3)

So, during 2019 Baker Corporation made a $57 investment in its current assets net of accounts payable and accruals.

Now we can substitute Baker Corporation's 2019 operating cash flow (OCF) of $392.3, its net fixed asset investment (NFAI) of $300, and its net current asset investment (NCAI) of $57 into Equation 4.4 to find its free cash flow (FCF):

$$FCF = \$392.3 - \$300 - \$57 = \$35.3$$

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.6 (3 of 3)

We can see that during 2019 Baker generated $35,300 of free cash flow, which it can use to pay its investors: creditors (payment of interest) and owners (payment of dividends). Thus, the firm generated adequate cash flow to cover all its operating costs and investments and had free cash flow available to pay investors. However, Baker's interest expense in 2019 was $70,000, so the firm is not generating enough FCF to provide a sufficient return to its investors.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.3 Cash Planning: Cash Budgets (1 of 7)

- Cash Budget (Cash Forecast)
  - A statement of the firm's planned inflows and outflows of cash that managers use to estimate its short-term cash requirements
- The Sales Forecast
  - The prediction of the firm's sales over a given period, based on external and/or internal data; used as the key input to the short-term financial planning process
  - External Forecast
    - A sales forecast based on the relationships observed between the firm's sales and certain key external economic indicators

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.3 Cash Planning: Cash Budgets (2 of 7)

- The Sales Forecast
  - Internal Forecast
    - A sales forecast based on a buildup, or consensus, of sales forecasts through the firm's own sales channels
- Preparing the Cash Budget
  - Total Cash Receipts
    - All of a firm's inflows of cash during a given financial period
  - Total Cash Disbursements
    - All outlays of cash by the firm during a given financial period

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.7 The General Format of the Cash Budget

	Jan.	Feb.	...	Nov.	Dec.
Total cash receipts	$XXA	$XXH		$XXN	$XXU
Less: Total cash disbursements	XXB	XXI	...	XXO	XXV
Net cash flow	$XXC	$XXJ		$XXP	$XXW
Add: Beginning cash	XXD	XXE	XXK	XXQ	XXR
Ending cash	$XXE	$XXK		$XXR	$XXX
Less: Minimum cash balance	XXF	XXL	...	XXS	XXY
Required total financing		$XXM		$XXT	
Excess cash balance	$XXG				$XXZ

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.8 A Schedule of Projected Cash Receipts for Coulson Industries ($000)

Sales forecast	Aug. $100	Sept. $200	Oct. $400	Nov. $300	Dec. $200
Cash sales (0.20)	$ 20	$ 40	$ 80	$ 60	$ 40
Collections of A/R:					
Lagged 1 month (0.50)		50	100	200	150
Lagged 2 months (0.30)			30	60	120
Other cash receipts					30
Total cash receipts	$ 20	$ 90	$ 210	$ 320	$ 340

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.9 A Schedule of Projected Cash Disbursements for Coulson Industries ($000)

Purchases (0.70 × sales)	Aug. $70	Sept. $140	Oct. $280	Nov. $210	Dec. $140
Cash purchases (0.10)	$ 7	$ 14	$ 28	$ 21	$ 14
Payments of A/P:					
Lagged 1 month (0.70)		49	98	196	147
Lagged 2 months (0.20)			14	28	56
Rent payments			5	5	5
Wages and salaries			48	38	28
Tax payments					25
Fixed-asset outlays				130	
Interest payments					10
Cash dividend payments			20		
Principal payments					20
Total cash disbursements	$ 7	$ 63	$ 213	$ 418	$ 305

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.7 (1 of 4)

Coulson Industries, a defense contractor, is developing a cash budget for October, November, and December. Coulson's sales in August and September were $100,000 and $200,000, respectively. Sales of $400,000, $300,000, and $200,000 have been forecast for October, November, and December, respectively. Historically, 20% of the firm's sales have been for cash, 50% have generated accounts receivable collected after 1 month, and the remaining 30% have generated accounts receivable collected after 2 months. Bad-debt expenses (uncollectible accounts) have been negligible. In December, the firm will receive a $30,000 dividend from stock in a subsidiary. The schedule of expected cash receipts for the company appears in Table 4.8. It contains the following:

Forecast sales Coulson expects sales to rise from August to October before falling slightly in November and December.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.7 (2 of 4)

Cash sales The cash sales shown for each month represent 20% of the total sales forecast for that month.

Collections of A/R These entries represent the collection of accounts receivable (A/R) resulting from sales in earlier months.

Lagged 1 month These figures represent sales made in the preceding month that generated accounts receivable collected in the current month. Because 50% of the current month's sales are collected 1 month later, the collections of A/R with a 1-month lag shown for September represent 50% of the sales in August, collections for October represent 50% of September sales, and so on.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.7 (3 of 4)

Lagged 2 months These figures represent sales made 2 months earlier that generated accounts receivable collected in the current month. Because 30% of sales are collected 2 months later, the collections with a 2-month lag shown for October represent 30% of the sales in August, and so on.

Other cash receipts These are cash receipts expected from sources other than sales. Interest received, dividends received, proceeds from the sale of equipment, stock and bond sale proceeds, and lease receipts may show up here. For Coulson Industries, the only other cash receipt is the $30,000 dividend due in December.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.7 (4 of 4)

Total cash receipts This figure represents the total of all the cash receipts listed for each month. For Coulson Industries, the figures for August and September represent actual cash inflows, whereas the figures for October, November, and December are forecasts based on expected sales and collections patterns.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.8 (1 of 3)

Coulson Industries has gathered the following data needed for preparing a cash disbursements schedule for October, November, and December.

Purchases The firm's purchases represent 70% of sales. Of this amount, Coulson pays 10% in cash up front. It pays 70% in the month immediately following the month of purchase, as highlighted in Table 4.9 by the row labeled *Lagged 1 month*. The company pays the remaining 20% 2 months following the month of purchase, as Table 4.9 shows in the row labeled *Lagged 2 months*.

Rent payments Coulson will pay rent of $5,000 each month.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.8 (2 of 3)

Wages and salaries Fixed salaries for the year are $96,000, or $8,000 per month. In addition, Coulson's financial managers estimate that wages equal 10% of monthly sales.

Tax payments Coulson must pay taxes of $25,000 in December.

Fixed-asset outlays The company will pay cash for new machinery costing $130,000 in November.

Interest payments An interest payment of $10,000 is due in December.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.8 (3 of 3)

Cash dividend payments Coulson will pay cash dividends of $20,000 in October.

Principal payments (loans) A $20,000 principal payment is due in December.

Repurchases or retirements of stock No repurchase or retirement of stock is expected between October and December.

Table 4.9 presents the firm's cash disbursements schedule, using the preceding data.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.3 Cash Planning: Cash Budgets (3 of 7)

- Preparing the Cash Budget
  - Net Cash Flow, Ending Cash, Financing, and Excess Cash
    - Net Cash Flow
      - The mathematical difference between the firm's cash receipts and its cash disbursements in each period
    - Ending Cash
      - The sum of the firm's beginning cash and its net cash flow for the period

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.3 Cash Planning: Cash Budgets (4 of 7)

- Preparing the Cash Budget
  - Net Cash Flow, Ending Cash, Financing, and Excess Cash
    - Required Total Financing
      - Amount of funds needed by the firm if the ending cash for the period is less than the desired minimum cash balance; typically represented by notes payable
    - Excess Cash Balance
      - The (excess) amount available for investment by the firm if the period's ending cash is greater than the desired minimum cash balance; assumed to be invested in marketable securities

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.9 (1 of 4)

Table 4.10 presents Coulson Industries' cash budget. The company wishes to maintain, as a reserve for unexpected needs, a minimum cash balance of $25,000. For Coulson Industries to keep this ending cash balance, it will need total borrowing of $76,000 in November and $41,000 in December. In October, the firm will have an excess cash balance of $22,000, which it can invest in an interest-earning marketable security. The required total financing figures in the cash budget refer to how much Coulson will owe at the end of the month; they do not represent the monthly changes in borrowing.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.9 (2 of 4)

We can easily find the monthly changes in borrowing and in excess cash by further analyzing the cash budget. In October, the $50,000 beginning cash, which becomes $47,000 after the $3,000 net cash outflow, results in a $22,000 excess cash balance (i.e., Coulson has $22,000 in cash above the $25,000 minimum). In November, Coulson has a net cash outflow of $98,000. From that we subtract the existing cash and marketable securities balance of $47,000 and find that the firm has a cash deficit of $51,000. Because Coulson wants a minimum of $25,000 of cash on hand at all times, in November the firm must borrow $76,000, enough to cover the cash deficit and replenish the cash balance to the desired minimum.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.9 (3 of 4)

In December, Coulson earns net cash inflows of $35,000, which it uses to repay some of its short-term borrowings, reducing the outstanding short-term debt from $76,000 to $41,000. In summary, the financial activities for each month would be as follows:

October: **Invest the $22,000** excess cash balance in marketable securities.

November: Liquidate the $22,000 of marketable securities and **borrow $76,000** (notes payable).

December: **Repay $35,000** of notes payable to leave $41,000 of outstanding required total financing.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.9 (4 of 4)

At the end of each of the 3 months, Coulson expects the following balances in cash, marketable securities, and notes payable:

	End-of-month balance ($000)		
Account	**Oct.**	**Nov.**	**Dec.**
Cash	$25	$25	$25
Marketable securities	22	0	0
Notes payable	0	76	41

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.10 A Cash Budget for Coulson Industries ($000)

	Oct.	Nov.	Dec.
Total cash receipts[a]	$210	$320	$340
Less: Total cash disbursements[b]	213	418	305
Net cash flow	–$ 3	–$ 98	$ 35
Add: Beginning cash	50	47	–51
Ending cash	$ 47	–$ 51	–$ 16
Less: Minimum cash balance	25	25	25
Required total financing (notes payable)[c]		$ 76	$ 41
Excess cash balance (marketable securities)[d]	$ 22		

[a]From Table 4.8.

[b]From Table 4.9.

[c]Values are placed in this line when the ending cash is less than the desired minimum cash balance. These amounts are typically financed short term and therefore are represented by notes payable.

[d]Values are placed in this line when the ending cash is greater than the desired minimum cash balance. These amounts are typically assumed to be invested short term and therefore are represented by marketable securities.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.3 Cash Planning: Cash Budgets (5 of 7)

- Evaluating the Cash Budget
  - The cash budget indicates whether managers should expect a cash shortage or surplus in each of the months covered by the forecast
  - It is a critical planning tool because it helps managers secure borrowing arrangements, such as bank lines of credit, before the firm actually needs the money

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Personal Finance Example 4.10 (1 of 3)

Individuals need to prepare budgets to make sure they can cover their current expenses (cash outflows) and save for the future. The personal budget is a short-term financial planning report that helps individuals or families achieve short-term financial goals. Personal budgets typically cover a 1-year period, broken into months.

A condensed version of a personal budget for the first quarter (3 months) is shown below.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Personal Finance Example 4.10 (2 of 3)

	Jan.	Feb.	Mar.
Income			
Take-home pay	$4,775	$4,775	$4,775
Investment income			90
(1) Total income	$4,775	$4,775	$4,865
Expenses			
(2) Total expenses	$4,026	$5,291	$7,396
Cash surplus or deficit [(1) – (2)]	$ 749	– $516	–$2,531
Cumulative cash surplus or deficit	$ 749	$ 233	–$2,298

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Personal Finance Example 4.10 (3 of 3)

The personal budget shows a cash surplus of $749 in January, followed by monthly deficits in February and March of $516 and $2,531, resulting in a cumulative deficit of $2,298 through March. Clearly, to cover the deficit, some action—such as increasing income, reducing expenses, drawing down savings, or borrowing—will be necessary to bring the budget into balance. Borrowing by using credit can offset a deficit in the short term but can lead to financial trouble if done repeatedly.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.3 Cash Planning: Cash Budgets (6 of 7)

- Coping with Uncertainty in the Cash Budget
  - Forecasts are almost inevitably wrong, at least to some degree
    - Scenario Analysis
      - Prepare several cash budgets, based on pessimistic, most likely, and optimistic forecasts
      - The use of several cash budgets, based on differing scenarios, also should give the financial manager a sense of the riskiness of various alternatives

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.11 (1 of 2)

Table 4.11 presents the summary of Coulson Industries' cash budget prepared for each month, using pessimistic, most likely, and optimistic estimates of total cash receipts and disbursements. The most likely estimate is based on the expected outcomes presented earlier.

During October, Coulson will, at worst, need $15,000 of financing and, at best, will have a $62,000 excess cash balance. During November, its financing requirement could be as high as $185,000, or it could experience an excess cash balance of $5,000. The December projections show maximum borrowing of $190,000, with a possible excess cash balance of $107,000.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.11 (2 of 2)

By considering extreme values in the pessimistic and optimistic outcomes, Coulson Industries should be better able to plan its cash requirements. For the 3-month period, the peak borrowing requirement under the worst circumstances would be $190,000, which happens to be considerably greater than the most likely estimate of $76,000.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.11 A Scenario Analysis of Coulson Industries' Cash Budget ($000)

	October			November			December		
	Pessimistic	Most likely	Optimistic	Pessimistic	Most likely	Optimistic	Pessimistic	Most likely	Optimistic
Total cash receipts	$ 160	$ 210	$ 285	$ 210	$ 320	$ 410	$ 275	$ 340	$ 422
Less: Total cash disbursements	200	213	248	380	418	467	280	305	320
Net cash flow	–$ 40	–$ 3	$ 37	–$ 170	–$ 98	–$ 57	–$ 5	$ 35	$ 102
Add: Beginning cash	50	50	50	10	47	87	–160	–51	30
Ending cash	$ 10	$ 47	$ 87	–$ 160	–$ 51	$ 30	–$ 165	–$ 16	$ 132
Less: Minimum cash balance	25	25	25	25	25	25	25	25	25
Required total financing	$ 15			$ 185	$ 76		$ 190	$ 41	
Excess cash balance		$ 22	$ 62			$ 5			$ 107

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.3 Cash Planning: Cash Budgets (7 of 7)

- Cash Flow within the Month
  - Because the cash budget shows cash flows only on a monthly basis, the information provided by the cash budget is not necessarily adequate for ensuring solvency
    - A firm must look more closely at its pattern of daily cash receipts and cash disbursements to ensure that adequate cash is available for paying bills as they come due

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.4 Profit Planning: Pro Forma Statements

- Pro Forma Statements
  - Projected, or forecast, income statements and balance sheets
- Preceding Year's Financial Statements
  - Starting point for pro forma financial statements
- Sales Forecast
  - The key input for pro forma statements

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.12 Fair Traders Income Statement for the Year Ended December 31, 2019

Sales revenue	
Mugs (1,000 units at $20/unit)	$ 20,000
Plates (2,000 units at $40/unit)	80,000
Total sales	$ 100,000
Less: Cost of goods sold	
Labor	$ 28,500
Materials	13,500
Overhead	38,000
Total cost of goods sold	$ 80,000
Gross profits	$ 20,000
Less: Operating expenses	10,000
Operating profits	$ 10,000
Less: Interest expense	1,000
Net profit before taxes	$ 9,000
Less: Taxes (assumed rate = 15%)	1,350
Net profit after taxes	$ 7,650
Less: Common stock dividends	4,000
To retained earnings	$ 3,650

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.13 Fair Traders Balance Sheet, December 31, 2019

Assets		**Liabilities and stockholders' equity**	
Cash	$ 6,000	Accounts payable	$ 7,000
Marketable securities	4,000	Taxes payable	300
Accounts receivable	13,000	Notes payable	8,300
Inventories	16,000	Other current liabilities	3,400
Total current assets	$ 39,000	Total current liabilities	$ 19,000
Net fixed assets	51,000	Long-term debt	18,000
Total assets	$ 90,000	Total liabilities	$ 37,000
		Common stock	30,000
		Retained earnings	23,000
		Total liabilities and stockholders' equity	$ 90,000

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.14 2020 Sales Forecast for Fair Traders

Unit sales		Dollar sales	
Mugs	1,500	Mugs ($25/unit)	$ 37,500
Plates	1,950	Plates ($50/unit)	97,500
		Total	$135,000

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.5 Preparing the Pro Forma Income Statement
(1 of 2)

- Percent of Sales Method
  - A simple method for developing the pro forma income statement; it forecasts sales and then expresses the various income statement items as percentages of projected sales
    - From Fair Traders' 2019 income statement:

$$\frac{\text{Cost of Goods Sold}}{\text{Sales}} = \frac{\$80{,}000}{\$100{,}000} = 0.800 = 80.0\%$$

$$\frac{\text{Operating Expenses}}{\text{Sales}} = \frac{\$10{,}000}{\$100{,}000} = 0.100 = 10.0\%$$

$$\frac{\text{Interest Expense}}{\text{Sales}} = \frac{\$1{,}000}{\$100{,}000} = 0.010 = 1.0\%$$

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.15 A Pro Forma Income Statement, Using the Percent-of-Sales Method, for Fair Traders for the Year Ended December 31, 2020

Sales revenue	$ 135,000
Less: Cost of goods sold (0.80)	108,000
Gross profits	$ 27,000
Less: Operating expenses (0.10)	13,500
Operating profits	$ 13,500
Less: Interest expense (0.01)	1,350
Net profit before taxes	$ 12,150
Less: Taxes (assumed rate = 15%)	1,823
Net profit after taxes	$ 10,327
Less: Common stock dividends	4,000
To retained earnings	$ 6,327

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.5 Preparing the Pro Forma Income Statement
(2 of 2)

- Considering Types of Costs and Expenses
  - The technique used to prepare the pro forma income statement in Table 4.15 assumes that all the firm's costs and expenses are variable
  - Because this approach assumes that all costs are variable, it may understate the increase in profits that will occur when sales increase
  - The best way to adjust for this when preparing a pro forma income statement is to break the firm's historical costs and expenses into fixed and variable components.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.12 (1 of 2)

Fair Traders' 2019 actual and 2020 pro forma income statements, broken into fixed and variable cost and expense components, follow:

Fair Traders Income Statements		
	2019 Actual	**2020 Pro Forma**
Sales revenue	$100,000	$135,000
Less: Cost of goods sold		
Fixed cost	40,000	40,000
Variable cost (0.40 × sales)	40,000	54,000
Gross profits	$ 20,000	$ 41,000
Less: Operating expenses		
Fixed expense	$ 5,000	$ 5,000
Variable expense (0.05 × sales)	5,000	6,750
Operating profits	$ 10,000	$ 29,250
Less: Interest expense (all fixed)	1,000	1,000
Net profit before taxes	$ 9,000	$ 28,250
Less: Taxes (assumed rate = 15%)	1,350	4,238
Net profit after taxes	$ 7,650	$ 24,012

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Example 4.12 (2 of 2)

Breaking Fair Traders' costs and expenses into fixed and variable components provides a more accurate projection of its pro forma profit. By assuming that *all* costs are variable (as shown in Table 4.15), we find that projected net profit before taxes would continue to equal 9% of sales (in 2019, $9,000 net profit before taxes ÷ $100,000 sales). Therefore, the 2020 net profit before taxes would have been $12,150 (0.09 × $135,000 projected sales) instead of the $28,250 obtained by using the firm's fixed-cost–variable-cost breakdown.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.6 Preparing the Pro Forma Balance Sheet (1 of 5)

- Judgmental Approach
  - A simplified approach for preparing the pro forma balance sheet under which the firm estimates the values of certain balance sheet accounts and uses its external financing as a balancing, or "plug," figure

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.6 Preparing the Pro Forma Balance Sheet (2 of 5)

- Assumptions for Fair Traders' 2020 Pro Forma Balance Sheet
  - A minimum cash balance of $6,000 is desired
  - Marketable securities will remain unchanged from their current level of $4,000
  - Accounts receivable on average represent about 45 days of sales (about 1/8 of a year)
    - Because Fair Traders' annual sales are projected to be $135,000, accounts receivable should average $16,875 (1/8 × $135,000)

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.6 Preparing the Pro Forma Balance Sheet (3 of 5)

- Assumptions for Fair Traders' 2020 Pro Forma Balance Sheet (continued)
  - The ending inventory should remain at a level of about $16,000, of which 25% should be raw materials and the remaining 75% should consist of finished goods
  - A new machine costing $20,000 will be purchased
    - Total depreciation for the year is $8,000
    - Adding the $20,000 acquisition to the existing net fixed assets of $51,000 and subtracting the depreciation of $8,000 yields net fixed assets of $63,000

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.6 Preparing the Pro Forma Balance Sheet (4 of 5)

- Assumptions for Fair Traders' 2020 Pro Forma Balance Sheet (continued)
  - Purchases will represent approximately 30% of annual sales, which in this case is approximately $40,500
    - The firm estimates that it can take 73 days on average to satisfy its accounts payable
    - Thus accounts payable should equal one-fifth (73 days ÷ 365 days) of the firm's purchases, or $8,100 (1/5 × $40,500)
  - Taxes payable will equal one-fourth of the current year's tax liability, which equals $455 (one-fourth of the tax liability of $1,823)

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.6 Preparing the Pro Forma Balance Sheet (5 of 5)

- Assumptions for Fair Traders' 2020 Pro Forma Balance Sheet (continued)
  - Notes payable will remain unchanged from their current level of $8,300
  - No change in other current liabilities is expected
  - The firm's long-term debt and its common stock will remain unchanged
  - Retained earnings will increase from the beginning level of $23,000 to $29,327
    - The increase of $6,327 represents the amount of retained earnings calculated in the year-end 2020 pro forma income statement

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.16 A Pro Forma Balance Sheet, Using the Judgmental Approach, for Fair Traders (December 31, 2020) (1 of 2)

Assets			**Liabilities and stockholders' equity**	
Cash		$6,000	Accounts payable	$ 8,100
Marketable securities		4,000	Taxes payable	455
Accounts receivable		16,875	Notes payable	8,300
Inventories			Other current liabilities	3,400
Raw materials	$ 4,000		Total current liabilities	$20,255
Finished goods	12,000		Long-term debt	18,000
Total inventory		16,000	Total liabilities	$38,255

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

Table 4.16 A Pro Forma Balance Sheet, Using the Judgmental Approach, for Fair Traders (December 31, 2020) (2 of 2)

Assets		**Liabilities and stockholders' equity**	
Total current assets	$ 42,875	Common stock	30,000
Net fixed assets	63,000	Retained earnings	29,327
Total assets	$ 105,875	Total	$ 97,582
		External financing required[a]	8,293
		Total liabilities and stockholders' equity	$105,875

[a]The amount of external financing needed to force the firm's balance sheet to balance. Because of the nature of the judgmental approach, the balance sheet is not expected to balance without some type of adjustment.

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.7 Evaluation of Pro Forma Statements (1 of 3)

- The major weaknesses of the approaches to pro forma statement development outlined above lie in two assumptions:
  - That the firm's past financial performance will be replicated in the future
  - That certain variables (such as cash, accounts receivable, and inventories) can be forced to take on certain "desired" values
- These assumptions cannot be justified solely on the basis of their ability to simplify the calculations involved

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.7 Evaluation of Pro Forma Statements (2 of 3)

- However pro forma statements are prepared, analysts must understand how to use them to make financial decisions
  - Financial managers and lenders can use pro forma statements to analyze the firm's inflows and outflows of cash, as well as its liquidity, activity, debt, profitability, and market value
  - Various ratios can be calculated from the pro forma income statement and balance sheet to evaluate performance

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.

4.7 Evaluation of Pro Forma Statements (3 of 3)

- However pro forma statements are prepared, analysts must understand how to use them to make financial decisions
  - Cash inflows and outflows can be evaluated by preparing a pro forma statement of cash flows
  - After analyzing the pro forma statements, the financial manager can take steps to adjust planned operations to achieve short-term financial goals

Pearson

Copyright © 2019 Pearson Education, Ltd. All Rights Reserved.